Dataset-Dependent Exploration–Exploitation Tradeoffs in Ligand-Based Virtual Screening

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Abstract

This project studies active learning for ligand-based virtual screening under a fixed evaluation budget. Rather than treating the task as standard classifier improvement, we focus on the practical objective of recovering active molecules as efficiently as possible. We compare pure exploitation, fixed exploration–exploitation mixtures, and a bandit-based adaptive strategy across five molecular screening datasets under scaffold split. To make the acquisition score more interpretable, we combine rank-transformed probability and uncertainty scores. To make cross-dataset evaluation fairer, we use normalized area under the screening curve (normalized AUSC) and enrichment factor (EF). The results show that pure exploitation is a strong baseline on most datasets. Exploration is often harmful, but a small amount of exploration can be beneficial in specific intermediate regimes, as seen on ClinTox where $\alpha = 0.1$ performs slightly better than $\alpha = 0$. Bandit adaptation is usually competitive with the best fixed strategy but does not consistently outperform it. These results suggest that the value of exploration is highly dataset-dependent and that strong exploitative baselines should not be ignored in virtual screening.

1 Introduction

Active learning is often motivated by the idea that querying uncertain examples improves downstream model quality. In ligand-based virtual screening, however, the practical objective is usually different: the goal is to recover active molecules as quickly as possible under a limited experimental budget. This creates a tension between *exploitation*, which prioritizes molecules predicted to be active, and *exploration*, which prioritizes uncertain molecules that may improve the model or reveal new regions of chemical space.

A common assumption is that some amount of exploration should be helpful. However, in screening tasks, uncertainty may not align with hit recovery, and a strong exploitation baseline may already be difficult to beat. In this project, we revisit this question using five molecular datasets: HIV, Tox21 NR-AR, ClinTox CT_TOX, BACE, and BBBP. We evaluate both fixed exploration weights and a bandit-based adaptive strategy.

Compared with the earlier version of the project, the current study makes three important methodological changes. First, we use *scaffold split* to make the task harder and more realistic than a random split. Second, we use a rank-based acquisition score so that the tradeoff parameter α has a cleaner interpretation. Third, we evaluate performance using normalized AUSC and EF so that comparisons across datasets with different prevalence are more meaningful.

2 Methods

2.1 Task formulation

We study pool-based active learning for binary molecular screening. At each iteration, a model is trained on the currently labeled set, scores the remaining unlabeled pool, and selects a batch of molecules for evaluation. The objective is not merely to improve classifier metrics, but to maximize active-molecule recovery under a fixed budget.

2.2 Model and representation

All molecules are represented using Morgan fingerprints, and the predictive model is a Random Forest classifier. The model outputs a probability score

$$P(y = 1 \mid x)$$

for each unlabeled molecule. Uncertainty is measured by binary entropy:

$$U(x) = -p \log p - (1 - p) \log(1 - p),$$

where $p = P(y = 1 \mid x)$.

2.3 Rank-based acquisition score

Because Random Forest probabilities are more reliable for ranking than for calibrated magnitude, and because probability and entropy have different numeric behavior, we transform both signals into within-round ranks. Let $\mathrm{rankP}(x)$ denote the rank-transformed predicted activity score and $\mathrm{rankU}(x)$ denote the rank-transformed uncertainty score. The acquisition function is

$$a_\alpha(x) = (1 - \alpha)\,\mathrm{rankP}(x) + \alpha\,\mathrm{rankU}(x).$$

Here:

- $\alpha = 0$ corresponds to pure exploitation,
- $\alpha = 1$ corresponds to pure uncertainty-driven exploration.

We evaluate

$$\alpha \in \{0,\ 0.1,\ 0.25,\ 0.5,\ 0.75,\ 1.0\}.$$

2.4 Bandit adaptive alpha

To avoid manually choosing α, we also test a bandit-based adaptive strategy. Each candidate α is treated as an arm, and the algorithm chooses an arm by an upper-confidence-bound (UCB) rule:

$$\mathrm{UCB}(\alpha_i) = \bar{r}_i + c\sqrt{\frac{\log t}{n_i}},$$

where $\bar{r}_i$ is the mean observed reward for arm α_i, n_i is the number of times it has been selected, and c controls exploration. The reward is the batch hit rate obtained after selecting a batch with that α.

2.5 Scaffold split

To make the task more difficult and reduce overly optimistic similarity-based generalization, we use scaffold split rather than random split. Molecules sharing the same Bemis–Murcko scaffold are grouped together, and these groups are assigned to pool and test partitions. This makes the held-out molecules more structurally distinct from the molecules used during active learning.

2.6 Metrics

Normalized AUSC. Let the screening curve plot cumulative recall as a function of the number of evaluated molecules. The area under this curve (AUSC) measures early and cumulative recovery. To reduce prevalence-driven distortion across datasets, we normalize AUSC by the expected AUSC of random sampling:

$$\text{NormAUSC} = \frac{\text{AUSC}_{\text{method}}}{\text{AUSC}_{\text{random}}}.$$

Enrichment Factor (EF). At the final budget k, enrichment factor is

$$\text{EF} = \frac{\text{hit rate in selected set}}{\text{overall prevalence}} = \frac{(\text{hits}/k)}{(A/N)},$$

where A is the total number of actives in the pool and N is the pool size.

These two metrics play complementary roles: normalized AUSC measures screening efficiency over the whole budget, while EF measures final concentration of active compounds relative to random sampling.

3 Experimental setup

We evaluate five datasets under the same active learning pipeline:

- HIV
- Tox21 (NR-AR)
- ClinTox (CT_TOX)
- BACE
- BBBP

For each dataset, we use scaffold split, an initial labeled set of 50 molecules, and a total screening budget of 500 molecules, queried in batches of 25. Results are averaged across multiple random seeds.

4 Results

4.1 Overall summary

Table **??** summarizes the main results. Pure exploitation ($\alpha = 0$) is best on HIV, Tox21, BACE, and BBBP. ClinTox is the only dataset for which a small amount of exploration improves performance slightly, with best fixed $\alpha = 0.1$. Bandit adaptation is generally competitive with the best fixed strategy, but does not consistently outperform it.

Dataset	$\alpha = 0$	Best fixed α	Bandit
HIV	11.72	11.72 ($\alpha = 0$)	11.16
Tox21	9.18	9.18 ($\alpha = 0$)	8.09
ClinTox	2.08	2.09 ($\alpha = 0.1$)	2.09
BACE	1.47	1.47 ($\alpha = 0$)	1.42
BBBP	1.29	1.29 ($\alpha = 0$)	1.21

Table 1: Normalized AUSC comparison. Best fixed α shown in parentheses.

Dataset	$\alpha = 0$	Best fixed α	Bandit
HIV	10.11	10.11	9.84
Tox21	6.41	6.41	6.35
ClinTox	1.84	1.82	1.83
BACE	1.50	1.50	1.47
BBBP	1.29	1.29	1.25

Table 2: Enrichment factor (EF) comparison.

4.2 Effect of fixed exploration weight

Figure 1 shows normalized AUSC as a function of fixed α, and Figure 2 shows the corresponding EF values.

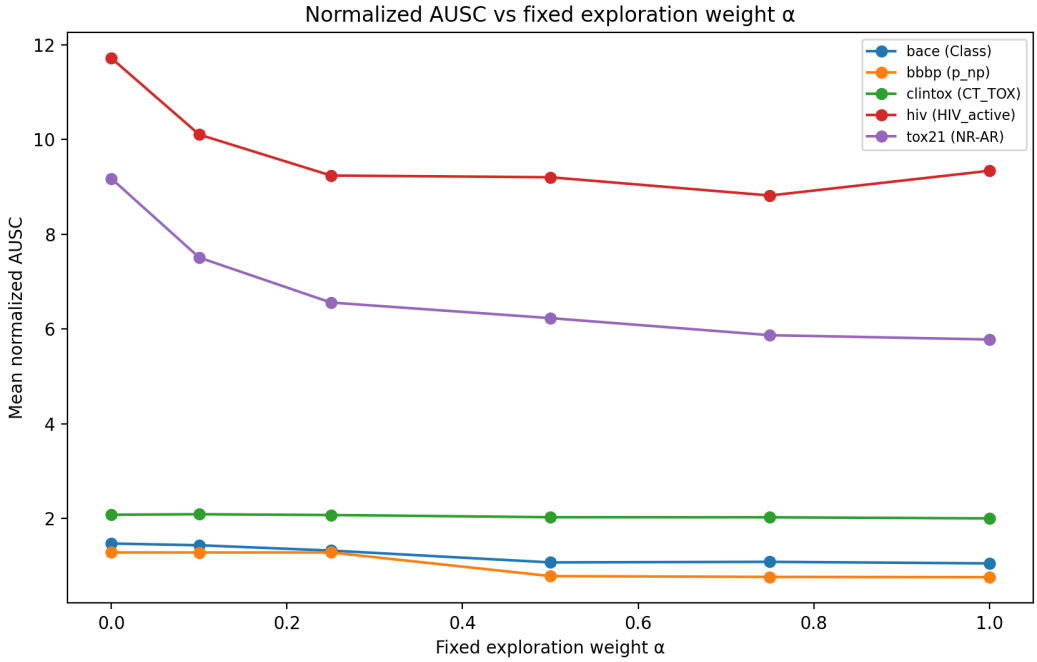

Figure 1: Normalized AUSC as a function of fixed exploration weight α. Most datasets favor pure exploitation, while ClinTox shows a slight benefit from small exploration.

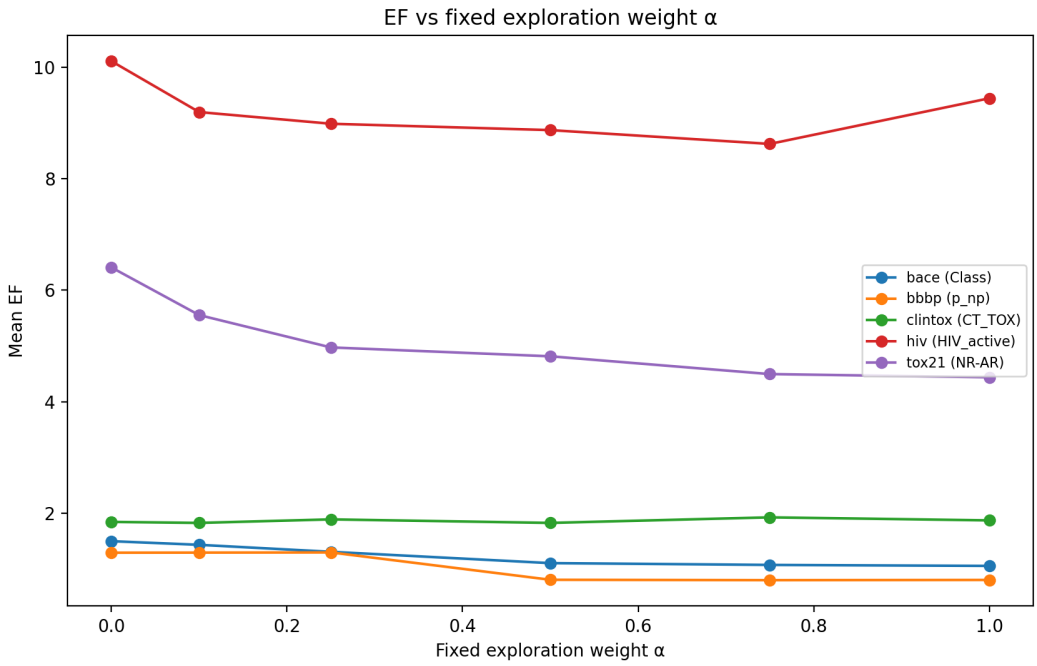

Figure 2: EF as a function of fixed exploration weight α. The same general pattern appears: exploration is often harmful, though ClinTox shows a small sweet spot at low α.

The fixed-alpha sweep reveals three broad regimes:

1. **Strong-signal regime**: On HIV and Tox21, performance declines clearly as α increases. This indicates that uncertainty-based exploration diverts budget away from highly ranked active candidates.

2. **Intermediate regime**: On ClinTox, a small amount of exploration ($\alpha = 0.1$) slightly improves normalized AUSC over pure exploitation, suggesting that modest exploration can help when model signal is present but not fully reliable.

3. **High-prevalence regime**: On BACE and BBBP, all methods are closer together, and the benefit of exploration is limited. Because actives are already relatively common, the potential gain from uncertainty-driven search is smaller.

4.3 Baseline, best fixed alpha, and bandit comparison

Figure 3 compares normalized AUSC for the baseline, best fixed alpha, and bandit. Figure 4 compares EF, and Figure 5 compares final hit counts.

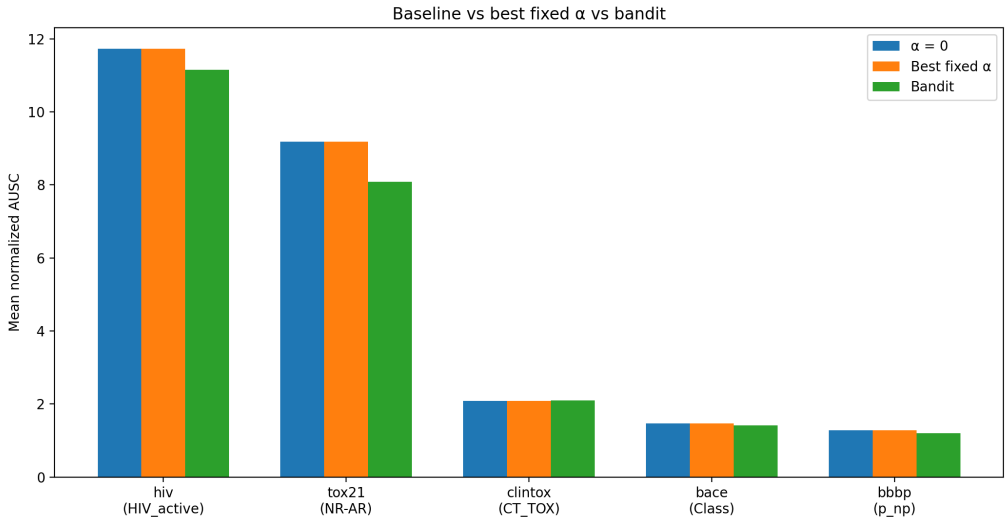

Figure 3: Comparison of normalized AUSC for pure exploitation, best fixed alpha, and bandit.

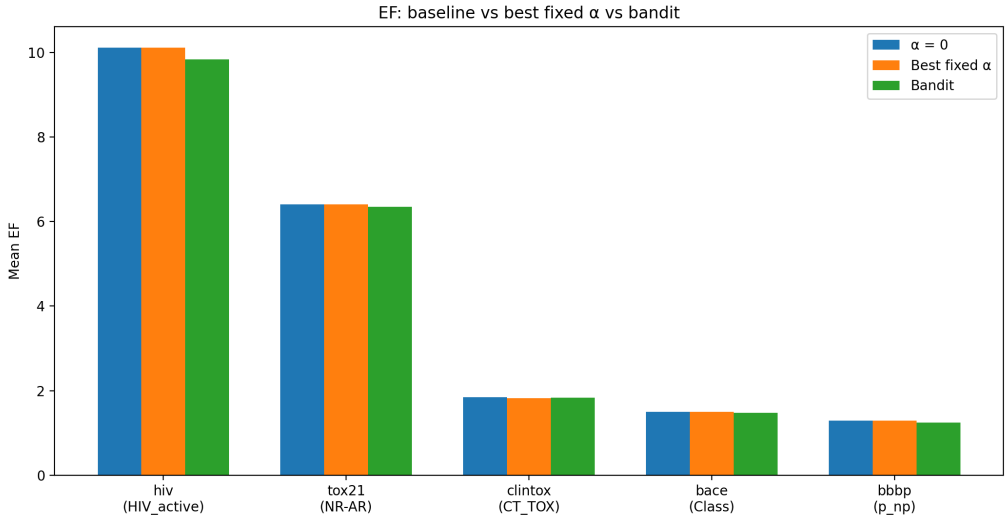

Figure 4: Comparison of EF for pure exploitation, best fixed alpha, and bandit.

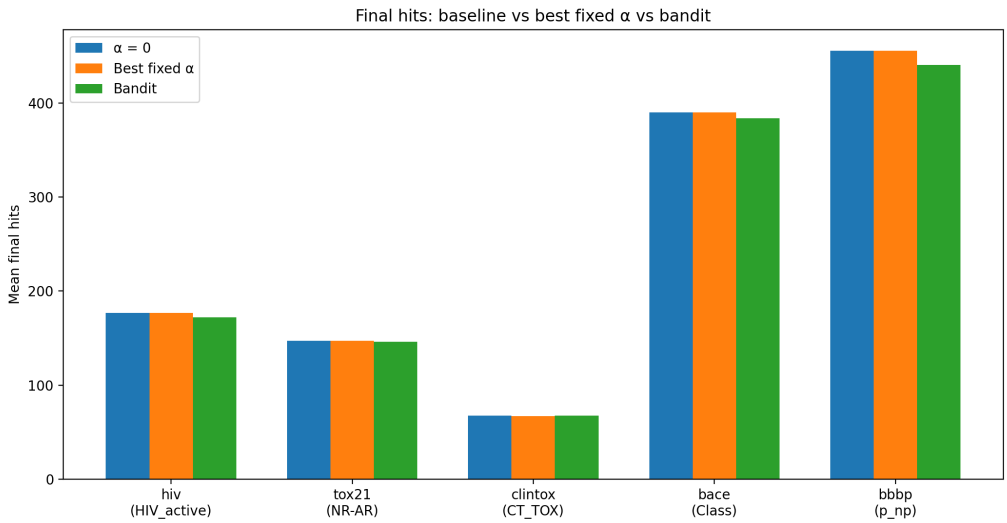

Figure 5: Comparison of final recovered active molecules for pure exploitation, best fixed alpha, and bandit.

Several patterns stand out. First, on four of the five datasets, the best fixed alpha is $\alpha = 0$, confirming that pure exploitation is already a strong baseline. Second, ClinTox is the only case where a nonzero fixed alpha performs best, and even there the gain is small. Third, bandit adaptation is often competitive but not decisively better. On Tox21, for example, bandit remains worse than $\alpha = 0$, which suggests that adaptively trying exploratory values can still waste screening budget when exploitation is already optimal. On ClinTox, by contrast, the bandit method is essentially tied with the best fixed alpha, indicating that adaptive control can be useful in intermediate regimes.

5 Discussion

The main conclusion of this project is that the value of exploration in virtual screening is strongly dataset-dependent. The naive expectation that uncertainty-driven exploration should generally help is not supported by these experiments.

Pure exploitation is a strong baseline. Across HIV, Tox21, BACE, and BBBP, pure exploitation performs best among the fixed-alpha strategies. This means that once the model has enough useful signal to rank candidate molecules, directly targeting predicted actives is often the most effective strategy for fast recovery.

Exploration is not universally beneficial. For several datasets, increasing α leads to monotonic degradation in both normalized AUSC and EF. In these cases, uncertainty appears to act more like noise than useful discovery pressure. This is especially clear on Tox21, where exploration steadily reduces performance.

Small exploration can help in intermediate regimes. ClinTox is the only dataset where a nonzero fixed alpha performs best, with $\alpha = 0.1$ yielding a slight improvement over $\alpha = 0$. This suggests that when model signal is present but imperfect, modest exploration can provide enough additional coverage or error correction to improve screening efficiency.

Bandit adaptation is competitive but not dominant. The bandit strategy usually avoids catastrophic choices and often comes close to the best fixed-alpha strategy. However, it does not consistently outperform a strong exploitation baseline. This is an important practical result: adaptive control may reduce the need for manual tuning, but it does not guarantee better hit recovery.

Metric choice matters. Using raw recall or raw AUSC alone can be misleading across datasets with very different prevalence. Normalized AUSC and EF make the cross-dataset comparison more interpretable and better aligned with the screening objective.

6 Conclusion

This study revisits the exploration–exploitation tradeoff in ligand-based virtual screening under a fixed evaluation budget. Using scaffold split, rank-based acquisition, normalized AUSC, and EF, we find that pure exploitation is a strong and difficult-to-beat baseline on most datasets. Exploration is often harmful, but not always: ClinTox shows that a small amount of exploration can be helpful in an intermediate regime. Bandit-based adaptive alpha selection is generally competitive, but it does not consistently outperform the best fixed strategy.

Overall, these results suggest that exploration should not be assumed to help automatically in active learning for virtual screening. Its benefit depends on dataset structure, signal quality, and task regime. A practical implication is that strong exploitative baselines should always be included, and that the usefulness of exploration should be evaluated empirically rather than assumed.

Future work

Several directions could strengthen this study:

- test additional uncertainty measures beyond entropy,
- evaluate other adaptive control rules,
- analyze larger-budget regimes,
- incorporate diversity-aware batch selection,
- and compare scaffold split with random split directly.